

Multi-Agent Large Language Model Tournaments for Lightweight Heuristic Generation in Augmented Reality Visual Acuity Screening

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Abstract. Traditional visual acuity testing is often inaccessible in resource limited settings due to its reliance on specialized environments and personnel. While augmented reality offers a portable alternative, current implementations overlook rich behavioral metadata by relying on basic linear calibrations. This study introduces a transparent, $O(1)$ complexity algorithm for converting AR-based Tumbling-E data to Snellen equivalents using a tournament-based Large Language Model (LLM) approach. Eight LLM agents competed iteratively via Elo ranking, with *gpt-oss-120b* generating interpretable logic and *Gemini 2.5 Flash* synthesizing executable code. Validated using eye-level stratified datasets (48 training/49 test right eyes; 48 training/47 test left eyes), the tournament-champion heuristic achieved a mean absolute error of 0.162 logMAR (right eyes) and 0.130 logMAR (left eyes), outperforming seven benchmark Machine Learning (ML) methods, including Transformer-based TabPFN and ML-based AutoGluon. Inference latency was minimized to 1.88 ms on CPU hardware, approximately $2,500\times$ faster than TabPFN (4,763 ms). One-way ANOVA showed significant differences among methods for right [F(2, 144) = 9.282, $p < 0.001$] and left [F(2, 139) = 12.351, $p < 0.001$] eyes. Tukey post-hoc analysis confirmed statistical equivalence to the ETDRS ground truth ($p = 0.718$ right; $p = 1.0$ left), whereas the baseline differed significantly ($p < 0.05$ right; $p < 0.001$ left). This approach yields an ultrafast, interpretable framework for automated visual acuity screening, broadening screening access on augmented reality hardware. The code is available at <https://github.com/FridayLover/MALT>.

Keywords: Visual Acuity, Augmented Reality, Multi-Agent LLM.

1 Introduction

Visual acuity (VA) is a fundamental metric for assessing vision-related quality of life, and VA impairment contributes substantially to the global burden of disease [14]. Large-scale epidemiological analyses identify uncorrected refractive error and cataract as leading causes of vision loss, particularly in low- and middle-income regions, where access to standardized clinical testing remains limited [19, 16]. Mobile health initiatives using smartphone-based charts have attempted to address this gap, but performance is degraded by uncontrolled testing conditions such as viewing distance, ambient illumination, and screen luminance [2].

Head-mounted augmented reality (AR) systems provide an alternative platform with fixed optical geometry and controlled luminance, enabling more standardized stimulus presentation across environments. Recent work on AR-based automated Tumbling-E testing has demonstrated strong agreement with standard clinical charts, supporting AR as a viable medium for VA assessment [17]. However, most existing approaches remain hardware-centric and require explicit calibration steps to convert pixel-level measurements into clinical Snellen or log-MAR units, which introduces implementation complexity [12].

In parallel, rapid advances in large language models (LLMs) have created new opportunities for transparent clinical decision support. Unlike conventional deep learning architectures for tabular data, which often function as black boxes, emerging agentic frameworks can generate interpretable reasoning chains and structured heuristics [22]. Techniques such as iterative self-refinement and multi-agent tournaments have proved effective for optimizing code and logical rules through feedback-driven evolution, enabling autonomous improvement without manual feature engineering [13, 8]. Adaptive response tuning and competitive evolutionary schemes further allow model behavior to be aligned with task-specific constraints [11].

Applying these agent-based optimization strategies to medical tabular data offers an avenue to derive lightweight, human-readable estimators that avoid the computational cost of transformer-based models such as TabPFN or complex AutoML ensembles such as AutoGluon [9, 7]. Within this context, the present work focuses on AR-derived VA testing data acquired using a Tumbling-E paradigm designed to emulate Early Treatment Diabetic Retinopathy Study (ETDRS) logic while remaining suitable for consumer-grade headsets [17].

The objectives of this study are as follows:

1. We develop a transparent heuristic framework for eye-level VA estimation in which the complete decision process is expressed through explicit, human-readable rules and equations.
2. We introduce a prompt-configured algorithm layer that supports task retargeting via high-level textual specifications.
3. We construct a streamlined, single-algorithm pipeline that yields near-instantaneous VA conversion for real-time AR interaction.
4. We design a resource-lean estimator that achieves competitive accuracy without GPU acceleration, supporting deployment on wearable devices in resource-constrained clinical and community settings.

2 Methods

2.1 Dataset

The dataset for this study was derived from a prior AR-based visual acuity work approved by the Institutional Review Board of the Faculty of Medicine, Chulalongkorn University, Bangkok, Thailand (Application No. 0629/66) [17]. A total of 109 adults aged ≥ 18 years were enrolled, with ETDRS visual acuity between 0.0 and 1.0 logMAR (Snellen 20/20–20/200), consistent with ICD-11 definitions of no, mild, and moderate distance vision impairment, and, after applying these criteria, analyzable data were available from 97 right eyes and 95 left eyes [17, 18]. The median age in the training cohort was 58.0–58.5 years, and 59.0 years in the test cohort for both eyes, and women accounted for approximately two-thirds of participants in each split (about 75% and 67%, respectively). Across both eyes, most samples were classified as having no impairment, with approximately 60% of right eyes and 65% of left eyes in the no-impairment category, around one-quarter in the mild-impairment range for each eye, and about 15% and 10% of right and left eyes, respectively, in the moderate-impairment range, while some participants had coexisting conditions such as cataract, glaucoma, or vitreoretinal disease that could degrade retinal image quality and potentially affect AR-based acuity measurements [17].

Visual acuity in each eligible eye was partitioned at the eye level into training and test sets, yielding 48 right and 48 left eyes for training and 49 right and 47 left eyes for testing, and we applied ICD-11–based distance-vision categories to stratify eyes across no, mild, and moderate impairment levels. Because each subject contributed at most one eye per side, no individual eye appeared in both training and test subsets for the same laterality, and all models were trained and evaluated independently for right and left eyes, preventing label leakage and ensuring that performance estimates for a given laterality were not biased by contralateral-eye inclusion in the opposite split.

2.2 Multi-Agent LLM Tournament Framework

The proposed framework was a transparent, tournament-based pipeline that employs multiple LLM agents to derive, implement, and iteratively refine visual acuity estimators from AR-derived tabular data. All clinical data remained on local hardware; only model prompts and generated code were processed by LLM agents, preserving data privacy and supporting reproducibility.

As illustrated in Fig. 1, the framework implemented a two-stage LLM pipeline. First, an open-weight gpt-oss-120B model in 4-bit quantized form generated candidate textual heuristics from structured task prompts [1]. A deterministic sampling temperature of 0.0 was used to minimize variance and reduce structural hallucinations, promoting stable algorithmic structure across generations [22]. Second, each heuristic was converted into executable Python code by Gemini 2.5 Flash, chosen for strong instruction adherence and efficient reasoning, again with a fixed temperature of 0.0 to ensure close fidelity to the proposed logic [4].

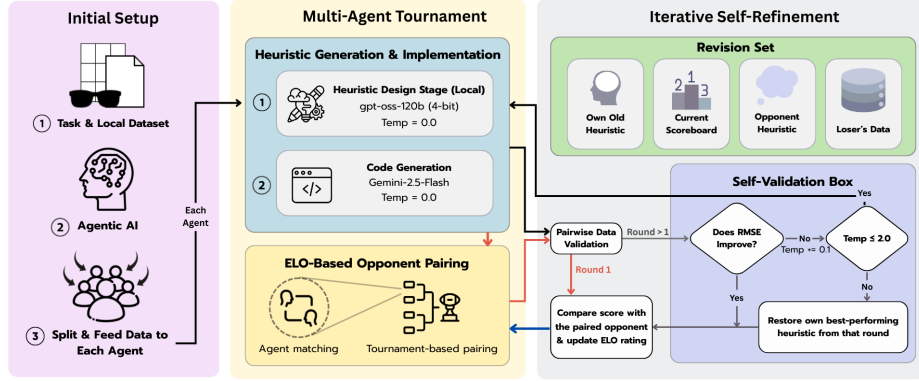


Fig. 1. Overview of the Multi-Agent LLM Tournament Framework. The pipeline used a two-stage generation process (Heuristic and Code Agents) with an iterative self-refinement loop governed by Elo-based ranking to evolve transparent visual acuity estimators.

To enable systematic refinement, the framework incorporated an internal validation-driven revision loop with temperature scheduling. The tournament initialized with eight contestant agents, each corresponding to a distinct heuristic generated under identical prompting conditions. Candidate heuristics are evaluated on held-out pairwise validation sets, and performance feedback was fed back into the generation process.

When a proposed revision failed to improve the best validation score, the sampling temperature for the heuristic-generation agent was increased by 0.1, up to a maximum of 2.0, to encourage controlled diversification of reasoning without destabilizing performance [22]. This process continued until an improvement was obtained or the maximum temperature was reached, with the overall tournament limited to three rounds in view of prior evidence that most gains in evolutionary frameworks occur during early iterations [8].

2.3 Prompt design

Three prompt templates are used to steer heuristic development: (1) an Initial Heuristic Generation Prompt, (2) a Code Generation Prompt, and (3) a Heuristic Revision Prompt. These prompts integrated self-reflection, explicit rationale generation, and comparative critique to structure reasoning, drawing on concepts from Tree-of-Thoughts and Self-Refine paradigms [13, 20].

The Initial Heuristic Generation Prompt instructed the model to decompose the solution into stepwise components with intermediate reasoning, analogous to thought decomposition in Tree-of-Thoughts approaches [20]. The Heuristic Revision Prompt adopted a Self-Refine structure, asking the agent to diagnose error patterns on validation data and to propose targeted modifications instead of regenerating an entirely new algorithm [13]. For reproducibility, the full prompt texts are provided in our code repository.

2.4 Tournament and Elo Rating

Heuristic contestants were ranked using an Elo-style rating system [6] with a baseline of 1200. For a pairwise match between agents i and j , the expected score E_i and updated rating R'_i were computed as:

$$E_i = \frac{1}{1 + 10^{(R_j - R_i)/400}}, \quad R'_i = R_i + K(\Delta) \cdot (S_i - E_i) \quad (1)$$

where R_i and R_j denote the current ratings, and $S_i \in \{0, 1\}$ is the actual match outcome.

Mirroring the Adaptive Response Tuning (ART) framework [11], the dynamic multiplier $K(\Delta)$ modulated updates based on the relative margin of victory Δ . Larger K -values were applied for substantial performance differences, emphasizing decisive wins and penalizing catastrophic failures, whereas smaller values preserved ranking stability for similar predictions [11]. Elo aggregation was preferred over raw win rates as it yielded a smoother, order-preserving summary of agent strength throughout the multi-round tournament [8, 11].

2.5 Model training and benchmarking

All models were evaluated using stratified 5-fold cross-validation at the eye level, with folds constructed to preserve the distribution of ICD-11 vision impairment grades within each split [18]. The proposed framework used internal self-competition on validation folds for heuristic selection, while external performance was reported on fixed eye-level test sets for right and left eyes.

Benchmark comparators included (1) an AR-based linear regression baseline that reimplemented the pixel-to-ETDRS transformation algorithms from the prior clinical AR study [17], (2) Support Vector Regression (SVR) to model potential non-linear relationships [5], (3) Gradient-Boosted Decision Tree methods composed of XGBoost, LightGBM, and CatBoost [3, 10, 15], and (4) AutoGluon-Tabular and TabPFN as modern AutoML and transformer-based baselines for tabular data [7, 9]. To match the deployment constraints of standalone AR headsets, all inference-time measurements were obtained on a CPU-only configuration without discrete GPU acceleration. Larger generic deep learning architectures were excluded to reduce overfitting risk, given the modest dataset size.

3 Experimental Results

3.1 Evolved heuristic structure

The Multi-Agent LLM Tournament produced explicit, human-readable algorithms for both eye cohorts, with full code available in our repository. The right-eye heuristic used streak-based effective accuracy with a global time multiplier, while the left-eye heuristic applied severity-weighted scoring with threshold-specific reaction-time penalties. Both models adjusted scores using reaction-time references treated as learned constants. Although parsing N trial lines required

an $O(N)$ pass to aggregate statistics, the core decision logic remained constant-depth over a bounded set of optotype sizes, yielding an effective per-eye evaluation cost of $O(1)$ with respect to the number of trials and aligning with the low-latency, CPU-only inference characteristics required for real-time AR-based screening.

3.2 Quantitative performance

Performance was evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) on the logMAR scale, along with the average inference time per subject. On the right-eye test set, the proposed heuristic outperformed all other methods, achieving the lowest error with an MAE of 0.162 and an RMSE of 0.212. TabPFN followed as the second-best performer with an MAE of 0.168.

Despite comparable accuracy, the proposed method maintained an extremely low computational footprint. Mean inference time was approximately 1.88 ms per subject, which closely matched the simple linear baseline (approximately 1.87 ms) and was markedly faster than AutoML and transformer-based comparators, with AutoGluon requiring approximately 189.6 ms and TabPFN about 4763.4 ms per subject on CPU-only hardware. The performance gap reflected architectural differences. While tree-based models traverse compiled decision structures, TabPFN processed each query jointly with training context through a large transformer network, which was efficient on accelerated hardware but imposed substantial latency on CPU-only devices.

The results for the left eye followed a consistent pattern. The proposed heuristic outperformed both the AR baseline and all machine learning regressors, achieving the superior scores of 0.130 logMAR (MAE) and 0.174 logMAR (RMSE). Furthermore, the heuristic demonstrated lower variance across cross-validation folds compared to TabPFN, suggesting greater stability within this cohort.

The evolutionary tournament required several hours to converge, whereas TabPFN effectively eliminated training time by relying on prior-data fitting. However, the heuristic training cost was incurred only once, after which deployment operates in constant time with minimal memory, prioritizing inference efficiency over training efficiency.

3.3 Statistical validation and agreement analysis

One-way ANOVA with Tukey post-hoc testing was used to compare predictions from the baseline model, the proposed heuristic, and clinical ground truth for each eye. For the right eye, ANOVA indicated significant differences among groups [$F(2, 144) = 9.282, p < 0.001$]. Tukey analysis showed that the baseline differed significantly from ground truth ($p < 0.05$), whereas predictions from the proposed model were not significantly different from ground truth ($p = 0.718$).

Left-eye analysis yielded similar results, with ANOVA demonstrating significant group differences [$F(2, 139) = 12.351, p < 0.001$]. The proposed model

Table 1. Performance comparison on Right Eye and Left Eye test sets. Values represent Mean $\pm$ SD in logMAR units. Inference time is averaged across cohorts.

Model	Right Eye (logMAR)		Left Eye (logMAR)		Time (ms)
	MAE	RMSE	MAE	RMSE	
Baseline [17]	0.208	0.260	0.200	0.256	1.87 ± 0.22
SVR [5]	0.206 ± 0.001	0.255 ± 0.003	0.161 ± 0.001	0.203 ± 0.001	1.96 ± 0.17
XGBoost [3]	0.184 ± 0.013	0.231 ± 0.009	0.172 ± 0.015	0.220 ± 0.021	2.89 ± 0.20
LightGBM [10]	0.236 ± 0.005	0.278 ± 0.004	0.180 ± 0.004	0.215 ± 0.004	2.24 ± 0.19
CatBoost [15]	0.166 ± 0.011	0.216 ± 0.013	0.145 ± 0.006	0.188 ± 0.009	2.51 ± 0.18
AutoGluon [7]	0.172 ± 0.012	0.223 ± 0.010	0.153 ± 0.011	0.195 ± 0.009	189.6 ± 31.2
TabPFN [9]	0.168 ± 0.010	0.217 ± 0.009	0.174 ± 0.017	0.215 ± 0.026	4763.4 ± 78.7
Proposed	0.162 ± 0.007	0.212 ± 0.013	0.130 ± 0.007	0.174 ± 0.006	1.88 ± 0.18

showed a statistically significant improvement over the baseline ($p < 0.001$) and remained statistically indistinguishable from the ground truth ($p = 1.0$). These findings indicated that the evolved heuristics approximated clinical VA measurements more closely than the original AR regression baseline [17].

Bland–Altman plots were generated to visualize agreement between predicted and measured logMAR acuity for both eyes (Fig. 2). For the right eye, the mean bias was 0.038 logMAR with limits of agreement (LoA) of approximately ± 0.370 logMAR, whereas the left eye showed a mean bias of -0.000 logMAR with LoA of approximately ± 0.324 logMAR. Although these LoAs are wider than the conventionally cited re-test variability of ± 0.15 logMAR, a recent systematic review reported that adult with an ocular comorbidity distance visual acuity LoA commonly range from approximately ± 0.10 to ± 0.49 logMAR, indicating that such variability is not unusual in clinical practice [21]. Within this empirically observed range, these limits were adequate for screening applications that required correct categorization into broad impairment strata, such as distinguishing mild from moderate impairment, rather than fine-grained management decisions. Overall, the Multi-Agent LLM Tournament framework produced algorithms that aligned closely with clinical ground truth while preserving the latency characteristics required for interactive AR-based screening.

4 Discussion and Conclusion

This study demonstrated that a tournament-based, multi-agent LLM framework could synthesize transparent heuristics for AR-derived visual acuity testing that reduced MAE and RMSE compared with the original AR regression baseline and achieved accuracy comparable to or better than SVR [5], Gradient-Boosted Trees [3, 10, 15], AutoGluon-Tabular [7], and TabPFN [9], while maintaining near-constant-time inference. From a clinical perspective, the combination of interpretability, low latency, and CPU-only deployment was particularly relevant for AR-based screening in primary or community care settings, where uncorrected refractive error and cataract were prevalent and access to standardized

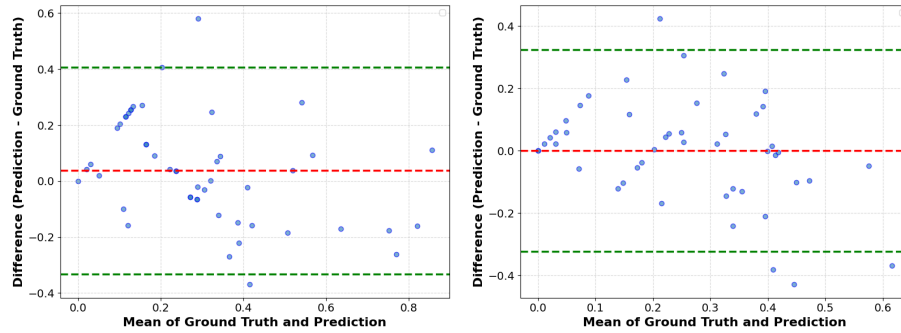


Fig. 2. Bland–Altman plots illustrating agreement between predicted values and ground truth for the Right Eye (left) and Left Eye (right). The dashed red line denotes the mean difference; dashed green lines indicate limits of agreement (± 1.96 SD).

visual acuity testing was often limited [14, 16, 17]. The explicit rule structure further allowed clinicians and engineers to inspect or adjust components such as reaction-time penalties and threshold criteria, supporting transparent adaptation to local workflows and constraints [22].

Beyond visual acuity, the framework illustrated a generalizable paradigm for converting black-box performance into human-readable, $O(1)$ decision rules on tabular data, leveraging prompt-configured agent layers and tournament-based refinement. In principle, the same strategy could be adapted to other ophthalmic endpoints, such as contrast sensitivity or visual field indices, provided that suitable tabular representations and evaluation metrics were available.

Several limitations should be mentioned. The current evaluation was based on a single AR visual acuity dataset with a modest sample size, and the learned heuristic was tailored to a specific Tumbling-E paradigm and display configuration, which might limit direct transfer to other devices or chart designs [17]. The evolutionary tournament also required several hours of offline computation and prospective performance in real-world screening workflows was not assessed in this study.

Future work should therefore include external validation on independent AR platforms, larger and more diverse cohorts, and broader impairment ranges, in line with recommendations for evaluating novel visual acuity technologies [17]. Prospective studies embedded in routine clinical or community screening programs could clarify usability, patient acceptability, and impact on referral patterns, complementing existing teleophthalmology initiatives targeting the global burden of avoidable vision impairment [14, 16]. Methodologically, further exploration of prompting strategies, reward functions, and tournament schedules may improve robustness and fairness across demographic and clinical subgroups, building on advances in self-refinement, Tree-of-Thoughts reasoning, and adaptive response tuning for LLMs [11, 13, 20].

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